

Myofascial force transmission to synergistic and antagonistic muscles

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Muscular force transmission

Muscular force transmission is central to bodily movement. Clearly, the tendon is the dedicated organ for the forces produced in the skeletal muscle to be transmitted to the bone to produce joint motion. Therefore, myotendinous force transmission is the key to muscular force transmission. However, force production in the muscle and transmission of the force to the joint are not independent processes. The muscle-tendon complex with its two mechanical elements arranged in series is exposed to forces coming from the joint, sharing them based on their relative stiffness. Consequently, for each muscle-tendon complex length, the muscle and the tendon attain certain lengths, and the length the muscle attains is a central determinant of muscular force production. This is because the global length of the muscle will affect the lengths of sarcomeres locally, which in turn is the key determinant for force production. Nevertheless, when it comes to local sarcomere lengths and particularly to what determines it mechanically, the picture is much more complex than just the question of how the muscle-tendon complex length is shared by the muscle and the tendon. In recent years, it has been shown in many studies that muscular force transmission is not limited to the tendon, but also involves muscle related connective tissues, that is, intramuscular fasciae (endo-, peri- and epimysium) as well as collagen-reinforced connective tissues connecting the muscle belly from

its epimysium to its surroundings, for example, the neurovascular tracts (Huijing, 2009; Wilke et al., 2018; Yucesoy, 2010). The important issue here is that the tendon is still the main component that will transmit the force to the joint to cause motion. However, the effect of considering those connective tissues as well is that the tendon's mechanical interaction with the muscle is not the only mechanism that will determine sarcomere lengths locally. Therefore, overall, muscular force transmission is a misnomer or represents an insufficient terminology. This is because, as argued above, myotendinous force transmission involves aspects of force production as well, but myofascial force transmission is more about force production than sole force transmission. This chapter will elaborate on these new viewpoints, aiming to provide movement practitioners with related background knowledge for their daily work with clients and patients.

Among other works by our research groups, an animal study published back in 2003 showed remarkable new scopes of muscle mechanics (Yucesoy et al., 2003). In that experiment, the isometric forces at both the proximal and distal tendons of the extensor digitorum longus (EDL) were measured simultaneously (Figure 14.1A) in two different conditions. In the first, the entire anterior crural compartment was kept intact. After lengthening the EDL distally, its distal forces, measured at different muscle lengths, were substantially higher (maximally by 46%)

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than the proximal forces. This proximo-distal force difference implies that part of the muscle's force may have been transmitted to other structures. This was supported by the second condition, in which the muscle's synergistic muscles, the tibialis anterior (TA) and the extensor hallucis longus (EHL), were removed. As a result, the EDL forces measured were different to those in the first condition although the testing protocol was identical (Figure 14.1B). At first sight, and considered together, these changes hence indicate the presence of force transmission between these synergistic muscles. Recall that, in general, the central mechanism of muscular force transmission to the joint involves the tendon. However, as the three muscles tested do not share a common tendon, the reported force transmission must be via the muscle-related connective tissue system. This mechanism has been referred to as myofascial force transmission (Yucesoy, 2010).

Note that the proximal forces of the EDL were lower in both conditions with such proximo-distal force differences being more pronounced in the intact condition. This means that the amplitude of muscle force produced at different lengths of the target muscle can change depending on the activity and positions of the surrounding muscles. Nevertheless, that was not the only remarkable effect shown. The EDL muscle's length range of force exertion was greater in the intact condition (Figure 14.1C). This means that the muscle length range between the muscle's active slack length (i.e. the shortest muscle length of non-zero muscle force production) and optimum length (i.e. the muscle length at which the peak force is measured) was manipulated. Therefore, the effect of inter-synergistic myofascial force transmission was not limited to integration of the force of one muscle into the force exerted at the tendon of another. Instead, the effect of inter-synergistic myofascial force

transmission is its capability to manipulate the mechanics of the target muscle substantially.

Muscle fiber-extracellular matrix integrity and mechanical interactions between these two domains

A condition-dependent varying output of a muscle as described above indicates altered force production, which can best be characterized with the mechanical interaction concept, that is, inter-synergistic myofascial force transmission is mechanical interaction between synergistic muscles. As lengths of sarcomeres of the muscle will determine the force the muscle can produce in every condition within which it functions, condition-dependent varying output of a muscle suggests that, despite being at an identical muscle length globally, the muscle's sarcomeres should attain differing lengths in different conditions. Note that sarcomere length is the result of a mechanical equilibrium between the forces acting on the sarcomere to change its length and the force the sarcomere produces to resist those. Sarcomeres with differing lengths within the muscle and even along the same muscle fibers will have differing force production. For this to take place, there must be forces acting on the sarcomere ends with varying amplitudes and directions. Such variable forces cannot be coming from the sarcomeres arranged in series within the same muscle fibers. However, in the level of a single muscle, they can come from the extracellular matrix (ECM) in the first place, and also from sarcomeres in adjacent muscle fibers, and act on the z-disks. Moreover, beyond the muscle level (compartment or even limb level, i.e. the muscle within the context of its intact connective tissue surroundings, which is the *in vivo* condition), these forces can also come from other myofascial structures outside the muscle, which are

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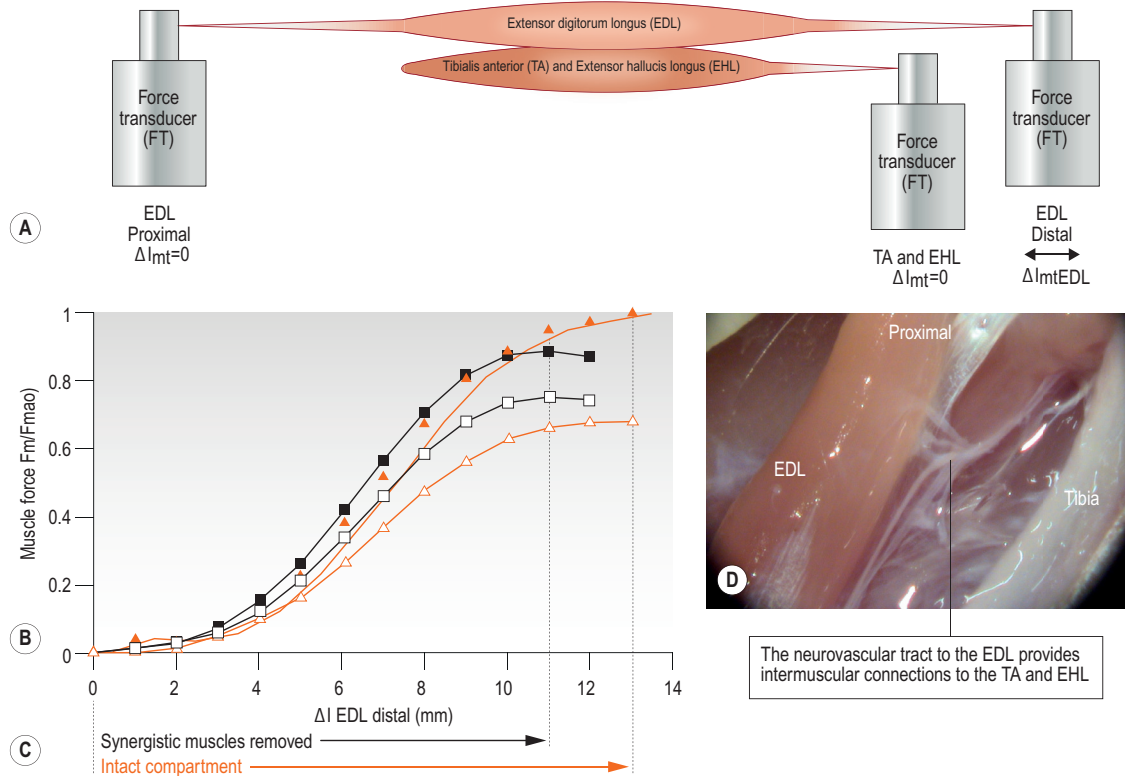


Figure 14.1 Assessment of the functional effects of intermuscular mechanical interactions

(A) In experiments performed for such a specific purpose, (1) the muscle belly was not dissected free from the surrounding structures, and instead of measurement of the muscle force only in one end and assuming identical force at the other end, (2) the forces exerted at both proximal and distal tendons of the target muscle were measured simultaneously using separate force transducers (FTs). The experiment discussed in the text involves anterior crural muscles. The forces of the EDL as the targeted muscle and the TA and the EHL were measured after activating these muscles maximally. Length changes of the EDL were imposed by changing the position of its distal force transducer (Δl_{mt} EDL), whereas the positions of the other FTs were kept constant ($\Delta l_{mt} = 0$). Further detailed illustrations and pictures of the experimental set-ups are available (Ateş and Yucesoğlu, 2014; Yucesoğlu and Ateş, 2018; Yucesoğlu et al., 2010a; 2012; 2015). **(B)** This approach was repeated under two conditions: (1) intact compartment and (2) after the synergistic muscles TA and EHL were removed. The results showed the characteristic effect: proximo-distal force differences (Yucesoğlu et al., 2003) quantified as a function of muscle length, in which changes indicate condition-dependent variable muscular force production. **(C)** Another remarkable effect is that the muscle's length range of force exertion is also condition-dependent instead of being a unique property. **(D)** The more proximal and stiffer part of the collagen neurovascular tract of the anterior crural compartment interconnects the EDL to the TA and EHL at their muscle belly and is continuous with compartmental connective tissues. These muscle-related connective tissues facilitate intermuscular mechanical interactions, which causes the changes in muscular mechanics, as illustrated in **(B)** and **(C)**.

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in continuity with the muscle's ECM. Therefore, if this is a tenable mechanism, inter-synergistic myofascial force transmission is not as simple as forces of other muscles being integrated to the tendon force of other muscles. Instead, it needs to be considered as mechanical interaction between synergistic muscles, which is characterized by a myofascial load concept (Yucesoy, 2010) with local sarcomere level effects translating into condition-dependent variable global output of the muscle to influence the joint moment and movement.

Tenability of that mechanism first requires presence of mechanical connections between the muscle fibers and the ECM. Although a certain general knowledge has existed for a long time that myofibers are connected to the ECM (Pond, 1982), this knowledge has not been implemented in physiological experiments. However, these two domains are connected mechanically along the full periphery of muscle fibers with complex structures. Inside the muscle fiber, internal myofibrils are interconnected via the protein desmin. On the other hand, the peripheral myofibrils are connected by trans-sarcolemmal molecules to the basal lamina (Berthier and Blaineau, 1997), which is connected to the endomysium. The endomysium forms a 3D structure of tunnels (Trotter and Purslow, 1992) within which the muscle fibers are operating (see, for an example illustration, Figure 1 in Yucesoy, 2010). A common simplistic consideration involves ascribing the mechanical connections between the muscle fibers and the endomysial tunnels only to the ends of the muscle fiber, where the myotendinous junctions are present. Mechanically, this is equivalent to considering muscle fiber as an uni-axially loaded member, which means that the only forces acting on the muscle fiber are located at the ends, and have identical amplitudes. Consequently, the length changes

along that member, that is, the lengths of sarcomeres arranged in series have been considered as identical. Another consequence of that has been to consider muscle fibers as units functioning independently of each other.

However, because of the trans-sarcolemmal connections, these viewpoints are not accurate hence tenable. Mechanical interactions between a muscle fiber and its epimysium as well as between several muscle fibers in a bundle (Street, 1983; Street and Ramsey, 1965), and the importance of the continuity of the ECM in intramuscular force transmission (Huijing, 1999) were shown in passive and active states. Therefore, muscle fibers cannot be considered as units functioning independently of each other. Additionally, the length of a sarcomere cannot be determined exclusively by its interaction with the sarcomeres arranged in series with it in the same muscle fiber. Instead, a more complex mechanical equilibrium, involving that and the forces exerted on the sarcomere by the fiber-reinforced ECM as well as the forces of the sarcomeres located in the neighboring muscle fibers, should be considered. Therefore, theoretically, a muscle fiber is a mechanical member with various loads acting on it along its length. Consequently, it is plausible that the forces acting at both ends of the muscle fiber are unequal and different parts of the muscle fiber may undergo unequal length changes. However, even this is unlikely to be the exhaustive consideration of muscle fiber-myofascial structure continuity and hence mechanical interactions.

Myofascial continuity between muscle and its surroundings

Intramuscular fasciae, including endomysium, perimysium and epimysium, are continuous within the muscle, but the epimysium also has

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connections to structures beyond the muscle. Within the same compartment, collagenous linkages between epimysia of adjacent muscles provide direct intermuscular connections, which can be considered as shared epimysia of those muscles. Additionally, with collagen-reinforced structures such as neurovascular tracts (Figure 14.1D), muscles within the same compartment and distant ones in other compartments, that is, antagonistic muscles, are interconnected. On the other hand, those myofascial structures are also continuous with intermuscular septa, compartmental fasciae, interosseous membranes, the periosteum and even ligaments. In certain locations, direct insertions of skeletal muscles into the myofascia are present. For example, the deltoides, latissimus dorsi and pectoralis muscles have fibrous expansions connecting to the brachial fascia (Stecco et al., 2008). In other locations, muscles originate from such fascial structures. The extensor hallucis longus muscle in the anterior crural compartment and the two muscles of the peroneal compartment originate from the anterior intermuscular septum. Note that these are antagonistic muscles. Structural continuity is not limited to the linkages between the fascia and the muscle. The connective tissue enveloping the adjacent compartmental muscles of the lower limb has been shown to fuse tightly with each other. In addition, myofascial continuity in an in-series arrangement is also present (Chapter 12). Therefore, via several different forms of structural continuity, synergistic and antagonistic muscles are mechanically integrated with each other. Taking into account such close morphological relationships between muscles and muscle-related fascial tissues, the effects of local stiffness changes and altered local muscular forces might affect both the tissue of origin and the surroundings. Conceivably, the insertions and origins of certain muscles onto and from myofascial structures allow for selective tensioning of the fascia,

which, in turn, supports and facilitates muscle activity via muscle and fascia mechanical interactions. Continuity of the lateral collateral ligament with fasciae suggests that muscle forces acting directly on the fascial system via muscle fiber attachments or indirectly via the continuity described above can manipulate not only muscle function, but also proprioception.

Effects of intermuscular mechanical interactions: Burden or fact with major functional implications?

Note that, using the terminology of muscular force transmission, the proximo-distal force differences addressed above comprise a solid example for the myofascial structures' force-transmitting capability. Yet this has sometimes been considered as unusual, even as a threat, in, for example, skeletal muscle physiology and musculoskeletal biomechanics communities. One common reason for this is because the proximo-distal force difference has been considered as a loss of some of the muscle's produced force being distracted away from the joint. However, due to the following reasons, alternative viewpoints can enable improvement of our understanding of skeletal muscle mechanics considerably.

First, this is not a realistic concern because all the connective tissues described above and, in particular, the neurovascular tracts, are vital elements of the machinery. They serve the absolutely necessary functions of accommodating or anchoring the force-producing motors and keeping them in physiological state. Additionally, they house the mechanoreceptors, so also serve the sensory aspect of the functioning of the motor. Therefore, those structures must be there, and intermuscular mechanical interactions attributable to them represent a fact to recognize rather than to ignore or fear.

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Second, the proximo-distal force difference needs to be considered as a mechanical mechanism determining the muscle's force production. This difference is the result of myofascial loads of possibly different amplitudes and directions acting on different locations of the epimysium. These loads arise from relative position changes of the muscle with respect to its surroundings, which occur due to: (1) differential length changes among muscles of the limb, for example, among mono- and bi-articular muscles, or among agonist and antagonistic muscles, (2) moment arm differences, also imposing differences in the length changes of several muscles. Such relative position changes cause the epimuscular myofascial connections of the muscles to become stretched and, in return, those structures exert myofascial loads on the muscle belly at the locations where they insert into the muscle, based on their non-linear mechanical properties and possible pre-strain (Yucesoy et al., 2005). Numerous finite element modeling studies have shown that such loads cause changes in sarcomere lengths and corresponding stress distributions along the muscle fibers, which is a metric for the muscle's force production (Yucesoy and Huijing, 2012; Turkoglu et al., 2014, Turkoglu; Yucesoy, 2016). Such stress values were shown to be in concert with the proximo-distal force differences, for example, lower values of stresses are calculated in the proximal segments of muscle fibers than in the distal segments in a muscle, which shows a smaller proximal force. This indicates that the proximo-distal force difference does not necessarily represent forces of other muscles to be integrated with the target muscle's force at the tendon, or otherwise some of the target muscle's forces leaking out to elsewhere. Instead, they represent myofascial loads acting on the target muscle, which affects the internal mechanics and modifies the muscle's force production.

However, another remarkable effect is that such mechanical interaction of the target muscle with its surroundings can yield changes to the shape of the muscle's length-force characteristics, including a change in the muscle's length range of force exertion, that is, its excursion (Yucesoy et al., 2003). This is determined by the heterogeneity of sarcomere lengths along, but more importantly, across different muscle fibers. A good metric for that is the mean sarcomere length of muscle fibers, the elevated heterogeneity of which across different muscle fibers was shown to shift the muscle's optimum length to a longer muscle length (Willems and Huijing, 1994), which means that the muscle's contribution to joint movement will be effective for a wider range of joint angles.

Muscle's effect at the joint moment and movement modified by its interaction with other structures in the compartment opens up new perspectives in musculoskeletal biomechanics as well as rehabilitation and sports science. That is why its extent beyond the compartment has become an issue of interest. In animal experiments involving multiple compartments, Huijing (2007) studied the interaction between the various lower leg muscles. For example, distal lengthening of the peronei muscles caused the active force of the TA to decrease by 25%. Such inter-antagonistic mechanical interaction was shown to affect the entire lower leg of the rat (Yucesoy et al., 2010b). Therefore, with a large body of evidence derived from animal studies and matching finite element modeling, the effects and mechanism of intermuscular mechanical interactions within the level of a whole limb has been revealed yielding epimuscular myofascial force transmission as a novel muscle mechanics paradigm. Note that the central component of this mechanism, that is, the heterogeneity of sarcomere lengths, has been shown by other groups as well (Moo et al., 2017), however, without addressing the

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cause of the heterogeneity or mentioning either intermuscular mechanical interactions or epimuscular myofascial force transmission.

Intermuscular mechanical interactions and sarcomere length inhomogeneity in human muscles *in vivo*

Effects of intermuscular mechanical interactions in terms of muscle belly displacements, stiffness or sarcomere length changes have been studied. Electrical percutaneous stimulation of the gastrocnemius was shown to cause displacement of the soleus suggesting presence of force transmission between these muscles (Bojsen-Møller et al., 2010). Also, Tian et al. (2012) studied those muscles' mechanical interaction and showed that knee flexion imposed at a constant ankle position affected soleus length. Although the source of such an effect could be myofascial force transmission, it could also be ascribed to the two muscles' mechanical interaction via their common Achilles tendon. Ultrasound shear wave elastography was used to assess the muscle shear modulus of the lower (Ateş et al., 2018a) and upper (Yoshitake et al., 2018) muscles. The former study showed mechanical interaction effects of the gastrocnemius lateral (GL) with the soleus, TA and peronei muscles in terms of altered shear modulus (an index of changes in muscle force) due to manipulated knee angle, despite the fact that the muscles do not cross the knee. The latter showed altered shear modulus effects between the biceps brachii and the brachialis, which was considered to isolate the myofascial channel as, in contrast to the gastrocnemius and the soleus, those muscles do not share a common tendon.

Magnetic resonance imaging (MRI) allows quantifying of local length changes of muscle tissue in human limb. Huijing et al. (2011) combined MRI analyses with a cadaver experiment for the

assessment of global length changes of the GM. This showed that local length changes caused by passively imposed knee motion at constant ankle angle are heterogeneous and the amplitudes of the global muscle length changes are much smaller. Note that Yaman et al. (2013) studied the remaining lower leg muscles, which do not cross the knee joint. Despite undergoing no global length changes, the local length changes shown, for example, for the TA, indicates effects of intermuscular mechanical interaction.

In particular, (Pamuk et al., 2016) and (Karakuzu et al., 2017) quantified length changes along GM muscle fibers as an indicator of sarcomere length changes. They used another MRI technique, that is, diffusion tensor imaging (DTI) to determine muscle fiber directions for that purpose. The former study showed that knee extension imposed in the passive state yielded strain heterogeneity along medial gastrocnemius (GM) fascicles. The latter study, conducted in the active state (15% of sustained maximal voluntary contraction), confirmed previous anticipations based on modeling that different parts of GM fascicles showed varying amplitudes and directions of length changes. Notably, Karakuzu et al. (2017) also specifically studied the neurovascular tracts and showed that the altered muscle position and activation yields local length changes within those structures. This was interesting in two ways: (1) straining of these collagen-reinforced tissues, interconnecting GM to other muscles implies their exposure to myofascial loads; (2) the locations where they insert into the muscle are the locations of high strain along muscle fiber length changes.

Therefore, MRI-DTI studies indicate that intermuscular mechanical interactions can lead to inhomogeneous length changes of sarcomeres along muscle fibers of human muscles *in vivo*. As addressed in the preceding sections, because

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this means that the muscle's contribution to joint moment and range of movement is not independent of other muscles' actions and positions, these techniques provide an important asset for our understanding of muscular mechanics. Note that evidence of variable sarcomere lengths across different regions of human muscle has been shown using other techniques such as microendoscopy (Lichtwark et al., 2018) with laser scanning microscopy (Llewellyn et al., 2008). This elegantly confirms the described mechanism of condition-dependent variable sarcomere lengths in general. However, again the resolution is very low, as only a few locations along selected fascicles can be studied. Note that some authors have questioned the relevance of epimuscular myofascial force transmission for physiologically representative or *in vivo* conditions (Maas, 2019). Although these are mostly animal studies, limited or trivial effects of such force transmission have also been reported, for example, altered knee angles imposing no effects on the soleus ankle moment in the rat (Tijs et al., 2015). It is important to consider the possibility of test conditions, joint position combinations or individuals showing marginal effects of intermuscular mechanical interactions. Nevertheless, even in the absence of such effects on the gross metrics, it may be hard to conclude that there are no local effects. After all, myofascial loads are supposed to impose no local effects only if the muscle is a rigid body, which is not true.

Implications for joint movement in health and disease

Pathological conditions involving impeded joint movement are highly conceivable implications for this mechanism. Cerebral palsy (CP) is one remarkable pathology, the etiology and treatment of which has been studied in the context of epimuscular myofascial force transmission.

This will be elaborated on here as an example of how this mechanism can affect muscle's contribution to joint moment and range of movement, and not specifically for the purpose of addressing pathological conditions. Instead, this can be considered as an overall phenomenon, which can be reflected to joint function in health and also to sport and other movement. In CP, limited joint mobility is ascribed to higher force production of spastic muscles typically in flexed joint positions and a narrow joint range of force production. Accordingly, it is considered that the muscle's peak force is encountered at shorter muscle lengths and it is incapable of producing active force at longer muscle lengths. However, intra-operative experiments showed that spastic gracilis (Ateş et al., 2013), semitendinosus (Ateş et al., 2016) and semi-membranosus (Yucesoy et al., 2017) show no profound force production in flexed knee positions and a narrow joint range of force exertion, if they are stimulated alone. This indicates the following: (1) spastic muscles do not necessarily have the assumed mechanics, and (2) the mechanical interactions between the target-activated spastic muscle and lower leg muscles in the passive state do not affect its mechanics. However, the simultaneous co-activation of knee flexor (Kaya et al., 2018) and extensor (Ateş et al., 2018b) muscles considerably elevated spastic semitendinosus forces. In gait relevant joint positions, such increases are highly relevant for the clinically observed limited knee extension (Kaya et al., 2019; 2020). Spastic gracilis also showed a shift of its peak force due to intermuscular mechanical interactions in the active state. Epimuscular myofascial force transmission should therefore be considered as a determinant of the pathological knee joint condition in CP, suggesting that tissue adaptations occur in the fascial structures rather than in spastic muscle's contractile apparatus. This also has major implications for

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surgical treatments and spasticity management using botulinum toxin. Muscle-lengthening surgery on a target muscle was shown to also affect other muscles, which were not interfered with because of their mutual mechanical interactions (Ateş et al., 2013). Botulinum toxin leads to stiffening of the exposed muscle's ECM. Via muscle fiber-ECM mechanical interactions, this leads to unintended contraindicated outcome of the treatment, including increased passive resistance and narrowed range of force production (Ateş and Yucesoy, 2014). Based on these studies, and leaving aside the abnormal neurological control in CP, the pathological condition may be formulated as mechanically abnormally linked muscles, which may not necessarily be abnormal themselves. Accordingly, the movement therapy is advised to focus on muscle-related connective tissues. On the other hand, overall, considering epimuscular myofascial force transmission as a determinant for muscle's force and movement production, suggests that this mechanism is also quite relevant for sports. This can, for example, be a determinant for the sportsman's talent for the particular branch of sports they are enrolled in. Co-activating an antagonistic muscle may lead to an extra force exerted at the joint to perform the particular task, or mechanical interactions between different muscles may be reflected as a widened joint range of movement. Such properties may help the athlete to reach the ball with appropriate joint positioning, and afterwards to exert sufficient force to either shoot it (e.g. in football after dribbling past a defender or after controlling a passed ball) or block it (e.g. in defense in basketball or volleyball, taking the necessary hand position and resisting the offensive action by avoiding a wrist joint extension or even applying a flexor moment) at the right moment. This is also likely to fine-tune the muscle's output for precision tasks such as playing a musical instrument. Additionally, the possible influence of

epimuscular myofascial force transmission on the mechanoreceptor function can be relevant. Proximo-distally exerted unequal forces as a central effect implies that Golgi tendon organs may sense force differences instead of just muscle force. Along muscle fiber length heterogeneity suggests that muscle spindles may sense local length changes rather than simply the muscle's length. Therefore, proprioception may be influenced by muscle-related connective tissues. A recent kinesio taping study indicated that the tape as a source of externally applied mechanical loading could have local effects within the entire limb (Pamuk and Yucesoy, 2015). This suggests that myofascial loads can affect other mechanoreceptors as well. Therefore, the sensory aspects of epimuscular myofascial force transmission may also be relevant both for improved movement performance as well as pain management. However, more involved investigations need to be conducted to achieve specific and clinically applicable knowledge.

In conclusion, data made available over the last 2 decades suggest that mechanical interactions between muscles and between muscle fibers and the ECM open new perspectives for understanding previously unknown aspects of muscular mechanics. This is paramount in identifying mechanisms of the etiology and treatment techniques of pathological conditions. Effects on local muscle tissue lengths are central to those. This may have a direct effect on the muscle's contribution to joint moment and movement, but even for joint positions in which muscle force is not directly affected, there may be effects on mechanoreceptors, sensory organs and sarcomeres. Therefore, the potential effects of such mechanical interactions on, for example, pain, joint position sense, proprioception, joint range of motion and precision tasks, which require dosing of the muscle's output, are highly plausible.

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